

AAPOR Report on Online Panels

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4. ERRORS OF NONOBSERVATION IN ONLINE PANEL SURVEYS

Any estimate of interest, whether a mean, a proportion, or a regression coefficient, is affected by sample-to-sample variations. These variations lead to some amount of imprecision (variance) in estimating the true parameters and this type of error is known as sampling error. In addition to sampling error, error of estimates (bias and variance) can result from the exclusion relevant types of respondents, a type of error known as nonobservation. Two types of nonobservation error, coverage and nonresponse, affect all modes of surveys, regardless of sample source, but have the potential to be more severe with online panels than with other types of surveys.

BASIC CONCEPTS REGARDING COVERAGE ISSUES IN ONLINE PANELS

The *target population* is the group of elements which the survey investigator wants to describe using the sample statistics. There are three important features of most target populations for sample surveys:

- Target populations are finite in size (i.e., at least theoretically, they can be counted).
- They have some time restrictions (i.e., they exist within a specified time frame).
- They are observable (i.e., they can be accessed).

For any survey, clearly specifying these aspects of target populations is desirable for clarity of the survey's purpose and for "replicability" of the survey. With regard to surveys using online panels, one potential target population is the full set of persons living in housing units in the United States during a specific time period. This target population would resemble that often chosen when other survey methods are used, such as RDD. In RDD, the target population might be altered to a smaller subset – those having access to a telephone (or more restrictively, access to a landline telephone). In Web surveys, the target population might be restricted to those having access to the Internet in their homes, at work, or at school.

A *housing unit* is a house, an apartment, a mobile home, a group of rooms, or a single room that is occupied (or if vacant, is intended for occupancy) as separate living quarters. Separate living quarters are those in which the occupants live and eat separately from any other people in the building and have direct access from the outside of the building or through a common hall. The occupants may be a single family, one person living alone, two or more families living together, or any other group of related or unrelated persons who share living arrangements. Not all people in the United States at any moment are adults; not all adults reside in housing units (e.g., some live in dormitories, prisons, long-term care medical facilities, or military barracks; still others are homeless). Online panels might choose a target population that includes those in such institutions.

Since the population changes over time, the time of the survey also defines the target population. Since many household surveys are conducted over a period of several days, weeks, or even months, and since the population is changing daily as persons move in and out of U.S. households, the target population of many household surveys is the set of persons in the household population during the survey

period. In practice, the members of households typically are “fixed” at the time of first contact in many surveys.

A *sampling frame* is a set of materials used to identify the elements of the target population. Sampling frames are lists or procedures intended to identify all elements of a target population. The frames may be maps of areas in which elements can be found, time periods during which target events would occur, or records in filing cabinets, among others. Sampling frames, at their simplest, consist of a list of population elements. As previously noted in Section 3, a sampling frame consisting of all Internet users or containing all possible U.S. email addresses does not exist.

All sampling frames have various features, each of which can affect the quality of sample survey estimates based on them: (1) undercoverage; (2) multiple mappings; and (3) duplication. *Coverage error* arises when such features produce systematic under- or over-estimation of target population parameters, or act to inflate the standard errors of estimates.

Undercoverage is the weakness of sampling frames prompting the greatest possibility of coverage error. It threatens to produce errors of nonobservation in survey statistics from failure to include parts of the target population in any survey using the frame. By definition, online panels seeking to represent the total U.S. population suffer from undercoverage because non-Internet users are not members. This is akin to the problem posed by cell-phone-only households for landline-only telephone surveys: Persons with only cell phones are not members of the landline frame (AAPOR, 2008).

Multiple mappings of frame to population (clustering) or population to frame (duplication) are problems in sample selection. For example, a telephone directory lists telephone households in order by surname, given name, and address. When sampling adults from this frame, an immediately obvious problem is the clustering of eligible persons that occurs. Clustering means that multiple elements (e.g., people) of the target population are represented by the same frame element (e.g., telephone number). A residential telephone listing in the directory may have a single adult or two or more adults living there. The same is true of email addresses which can be shared by persons (e.g., smithfamily257@comcast.net). And so there can be ambiguity about which person connected to that address is being asked to participate. However, many online panels, as a matter of policy, insist that each member have his or her own email address, unshared with another person.

Duplication means that a single target population element (e.g., a person) is associated with multiple frame elements (e.g., multiple telephone numbers – business, cell, home, etc.). The problem that arises with this kind of frame problem is similar to that encountered with clustering. Target population elements with multiple frame units have higher chances of selection and will be overrepresented in the sample, relative to the population. If there is a correlation between duplication and the survey variables of interest, then the estimates of the survey measures will be biased. In survey measure estimation, the problem is that both the presence of duplication and the correlation between duplication and survey variables are often unknown. In the case of online panels, different email addresses may be held by the same person. While panel companies attempt to enforce a one email/one member rule at the joining stage, the effectiveness of those measures is unclear. In addition, it is not unusual for a member of one panel to also be a member of other panels. Surveys that draw on multiple panels therefore run the risk of sampling the same person multiple times. A recent study by The Advertising Research Foundation that conducted the same survey across 17 different panel companies found a duplication rate of 40 percent or 16 percent, depending on how it is measured (Walker, Pettit, and Robinson, 2009). In general, the more

that large samples from different panels are combined for a given study, the greater the risk of respondent duplication.

ONLINE PANEL SURVEYS, FRAMES, AND COVERAGE ISSUES

Now we can apply more specifically the general notions of target populations, sampling frames, and coverage issues to online panel surveys. As previously noted, there is no full and complete list of email addresses that can be used as a sampling frame for general population Web surveys. Even if such a frame existed, it would fail to cover a significant portion of the U.S. adult population, a common target population of interest to commercial and social researchers. It would have duplication problems because a person can have more than one email address. It would have clustering problems because more than one person can share an email address.

As described in Section 3, volunteer panels do not attempt to build a complete sampling frame of email addresses connected to persons. Their approach differs from classic sampling in a number of ways.

First, often the entire notion of a sample frame is skipped. Instead, the panel company focuses on the recruitment and sampling steps. Persons with Internet access are solicited in a wide variety of ways to acquire as diverse and large a group as possible.

Second, a common evaluative criterion of a volunteer panel is not full coverage of the household population but the collection of a set of persons with sufficient diversity on attributes related to the type of surveys the panel supports. For a panel built to support research on household products, this might include diversity on income, household size, ethnic sub-cultural identity, age, and various other lifestyle attributes. For a panel focused on public opinion, the diversity of interest might be income, age, gender, ethnicity, party identification, and social engagement. The only way to evaluate whether the panel has the desired diversity is to compare the assembly of volunteers to the full target population. To do this most thoroughly would require census data on all variables, and an assessment of means, variances, and covariances for all combinations of variables. Even then, there is no statistical theory that would offer assurance that some other variable not assessed would have the desired diversity.

Third, within the constraints of lockout periods, online panels can repeatedly sample from the same set of assembled willing survey participants and will send survey solicitations to a member as long as the member responds. There generally is little attention to systematically reflecting dynamic change in the full population of person-level email addresses. To date, the practice of systematically rotating the sample so that new entrants to the target population are properly represented is seldom used.

In short, without a universal frame of email addresses with known links to individual population elements, some panel practices will ignore the frame development step. Without a well-defined sampling frame, the coverage error of resulting estimates is unknowable.

One might argue that lists built using volunteers could be said to be frames and are sometimes used as such. However, the nature of these frames versus probability sample frames is quite different. The goal is usually to make inferences, not to the artificial population of panel members, but to the broader population of U.S. households or adults.

Further, defining coverage for panels is not quite as straightforward as defining coverage for other modes such as telephone or face-to-face frames. Every adult in a household with a telephone is

considered to be covered by the frame, as is every adult in an occupied dwelling unit selected for an area probability sample. Although people may have access to the Internet, either at home or at some other location, not everyone in a household may actually use the Internet.

In a household with Internet access and at least one user, non-users in the household could be considered as covered since the user can either answer for the other adults or bring another adult to the computer in the same way as the person who answers the telephone or is interviewed in person has the opportunity to engage other members of the household in the interview process. Individuals who only have Internet access at a location other than at home may or may not be considered as being covered. For example, an individual's access at work may be limited by corporate policy to business-related activities.

As we noted in the previous section, estimates of access to the Internet can vary widely. Mediamark Research and Intelligence (MRI) in their in-person *Survey of the American Consumer*TM (PiekarSKI et al, 2008) estimated that in 2008 approximately 85 percent of the adult population of the continental U.S. has *some* Internet access. This is considerably higher than the previously-cited 77 percent reported by the 2009 CPS and the 74 percent reported by Pew. The demographics of those without Internet access in some cases differ significantly from those with access and these differences may persist even if statistical adjustments are made. Adults without Internet access are more than twice as likely to be over the age of 65 as the general adult population. Those without Internet access are also more likely to be a member of a minority group, have incomes less than \$25,000, have a high school education or less, be unemployed or retired, not own their home, live in rural counties, or in the South Census Region.

Since participating in an online survey or belonging to an online panel is probably limited to those individuals who actually use the Internet, it is reasonable to assume that *coverage* really means using the Internet. The MRI, CPS, and Pew estimates are all lower than the household (88 percent) and population (89 percent) telephone coverage estimates from 1970 that led to the acceptability of using telephone surveys in place of in-person surveys.⁵

The foregoing discussion has focused on the interplay of sample frames and coverage issues within the context of the recruiting practices commonly employed for volunteer panels. Of course, email addresses are not the only form of sampling frame for Web surveys. As described in Section 3, people are sometimes recruited by telephone or mail and asked to join an online panel. In such a design, the coverage issues are those that pertain to the sampling frame used (e.g., RDD). Panels recruited in this manner often try to ameliorate the coverage problems deriving from not everyone contacted having Internet access either by providing the needed equipment and access or by conducting surveys with this subgroup using offline methods (telephone or mail). The extent to which providing Internet access might change the behaviors and attitudes of panel members remains unknown.

UNIT NONRESPONSE AND NONRESPONSE ERROR

⁵ We note that due to the emergence of cell-only households landline coverage has dropped significantly in recent years and is now less than 80 percent and may soon approach or even be less than household Internet penetration. However, it also is becoming common practice to include cell phone samples in telephone studies that aim to represent the full population.

Unit nonresponse, in contrast to coverage of the target population, concerns the failure to measure a unit in a sample. Unit nonresponse occurs when a person selected for a sample does not respond to the survey. This is distinguished from *item nonresponse*, which occurs when a respondent skips a question within a survey – either intentionally or unintentionally. In our treatment of nonresponse in this section we are referring to unit nonresponse. In traditional survey designs, this is a nonobservation error that arises after the sampling step from a sampling frame that covers a given target population. Unlike online panels that are established using probability sampling methods, volunteer panels are not established using probabilistic sampling techniques. This in turn affects various aspects of how nonresponse and nonresponse bias in such panels are conceptualized and measured, and what strategies are effective in trying to reduce these problems.

There are four stages in the development, use, and management of volunteer panels where nonresponse can become an issue: (1) recruitment; (2) joining and profiling; (3) specific study sampling; and (4) panel maintenance.

RECRUITMENT STAGE. As described in Section 3, none of the means by which nonprobability online panel members are recruited allow those who are establishing and managing the panel to know with any certainty from what base (i.e., target population) their volunteer members come. Because of this, there is no way of knowing anything precise about the size or nature of the nonresponse that occurs at the recruitment stage. However, the size of the nonresponse is very likely to be considerable. Similarly, the nature of the nonresponse may not be random with respect to variables of interest to health, social science, consumer and general marketing researchers. The latter follows from the known demographics of those who enroll as members, and their differences from the general population. The motives that lead some who are exposed to invitations to join online panels often reflect attitudinal and personality differences between them and others in the general population (e.g., Bosnjak, Tuten, and Wittman 2005). Psychographic variables (attitudes and personality measures of individuals) are starting to be used by online panel companies in their efforts to create panels and samples that are potentially more representative of the larger population.

Empirical evaluations of online panels abroad and in the U.S. leave no doubt that those who choose to join online panels differ in important and nonignorable ways from those who do not. For example, researchers directing the Dutch online panel comparison study (Vonk, van Ossenbruggen, and Willems, 2006) report that ethnic minorities and immigrant groups are systematically underrepresented in Dutch panels. They also found that, relative to the general population, online panels contained disproportionately more voters, more Socialist Party supporters, more heavy Internet users, and fewer church-goers.

Similarly, researchers in the U.S. have documented that online panels are disproportionately comprised of whites, more active Internet users, and those with higher levels of educational attainment (Couper, 2000; Dever et al., 2008; Chang and Krosnick, 2009; and Malhotra and Krosnick, 2007). In other words, the membership of a panel generally reflects the demographic bias in Internet use. Attitudinal and behavioral differences similar to those reported by Vonk et al. also exist for Internet users in the U.S. based on an analysis of the online population conducted by Piekarski, et al. (2008) using data from the MRI in-person *Survey of the American Consumer*TM. After standard demographic weighting, U.S. Internet users (any use) who reported being online five or more times per day were found to be considerably more involved in civic and political activities than the general U.S. population. The researchers also found that frequent users in the U.S. placed less importance on religion, traditional

gender roles and more importance on environmental issues. In this study, panel members reported even greater differences from the general population for these activities when the unweighted data were examined.

If online panel members belonging to under-represented groups are similar to group members who are not in the panel, then the risk of bias is diminished under an appropriate adjustment procedure. However, there is evidence to suggest that such within-group homogeneity may be a poor assumption. In the Dutch online panel comparison study, some 62 percent of respondents were members of multiple panels. The mean number of memberships for all respondents was 2.7 panels. Frequent participation in online surveys does not necessarily mean that an individual will be less representative of a certain group than he or she would otherwise be, but panel members clearly differed on this activity. Vonk et al. (2006) concluded that “panels comprise a specific group of respondents that differ on relevant criteria from the national population. The representative power of online panels is more limited than assumed so far.” Based on their analyses, the Dutch panels also had a number of differences between them, much like the house effects we have seen in RDD telephone samples (Converse and Traugott, 1986).

JOINING AND PROFILING STAGES. As described in Section 3, many panels require individuals wishing to enroll first to indicate their willingness join by clicking through to the panel company’s registration page and entering some personal information, typically their email address and key demographics (minimally age to ensure “age of consent”). Then he or she typically is sent an email to which the volunteer must respond to indicate that he/she in fact did sign up for the panel. This two-step process constitutes the “double opt-in” the vast majority of online panels require before someone is officially recognized as a member and available for specific studies. Potential members who initially enroll may choose not to complete the profiling questionnaire. Alvarez et al. (2003) report that just over 6 percent of those who clicked through a banner ad to the panel registration page eventually completed all the steps required to become a panel member. Those building and managing online panels can learn something about this nonresponse by comparing the limited data about the prospective panel member gathered at the recruitment stage between those who complete the profiling stage versus those who do not. However, very little has been reported.

SPECIFIC STUDY STAGE. Once a person has joined the panel, he or she likely will be selected as one of the sampled panel members invited to participate in specific surveys. There are several reasons why a sampled member may not end up participating fully or at all in a specific survey. These include:

- Refusal due to any number of reasons such as lack of interest, survey length, or a heavy volume of survey invitations;
- Failure to qualify due either to not meeting the study’s eligibility criteria or not completing within the survey’s defined field period;
- Technical problems that prevent either delivery of the survey invitation or access to and completion of the online questionnaire.

Those building and managing online panels can learn a great deal about nonresponse at this stage by using the extensive data about their panel members gathered at the initial recruitment and profiling stages, or from information gleaned from any previous surveys the members may have completed as panel members. Analyses can compare those sampled members who complete the specific survey questionnaire with those panel members who were sampled for the specific survey but did not complete the

questionnaire. It is not unusual to find that response rates vary across subgroups that are of analytic interest, particularly demographic groups. For telephone, mail, and face-to-face surveys, nonresponse has often been reported to be higher among those less educated, older, less affluent, or male (Dillman, 1978; Suchman and McCandless, 1940; Wardle, Robb, and Johnson, 2002). The pattern for nonprobability panels may be somewhat different – in one study, nonresponse was higher among panel members who were elderly, racial or ethnic minorities, unmarried, less educated, or highly affluent (Knapton and Myers, 2005). Gender was found to be unrelated to nonresponse in the online panel they studied. Different panels can vary substantially in their composition and in their recruitment and maintenance strategies, so the results from this one study may not generalize to other panels. Despite a great deal of data being available to investigate this issue, little has yet been publically reported.

Some panel companies attempt to address differential nonresponse at the sampling stage, i.e., before data collection even begins. In theory, one can achieve a final responding sample that is balanced on the characteristics of interest by disproportionately sampling panel members belonging to historically low response rate groups at higher rates. For example, Hispanic panel members might be sampled for a specific survey at a higher rate than other members in anticipation of disproportionately more Hispanics not responding. Granted, this sampling approach is possible outside of the online panel context, but it requires certain information (e.g., demographics) about units on the frame – information that is also often available in other survey designs (certain telephone exchanges and blocks of numbers may have a high density of Hispanics or blacks, or be more likely to be higher or lower income households; demographics have often been linked with zip codes so they can be used extensively in sampling). Bethlehem and Stoop (2007) note that this practice of *preemptive differential nonresponse adjustment* is becoming more challenging as Web survey response rates decline. Control over the final composition can only be achieved by taking into account differential response propensities of many different groups, using information about response behavior from previous, similar surveys. *However, even when balance is achieved on the desired dimensions, there is no guarantee that nonresponse error has been eliminated or even reduced.* The success of the technique relies on the assumption that nonresponding panel members within specific groups are similar to respondents within the same specific groups on the measures of interest.

PANEL MAINTENANCE AND PANEL ATTRITION. As mentioned earlier, excessive attrition can be a problem for replicating results within a panel. There are two types of panel attrition: *forced* and *normal*. Some panels may set a fixed length of time (e.g., 24 months, 100 surveys) during which a person can remain a panel member, but many do not. Forced attrition occurs within panels that have a maximum duration of elapsed time or a threshold number of surveys for which any one member can remain in the panel. After that criterion has been reached the member is removed (dropped) from the panel. Whether that person is eligible to rejoin the panel after some passage of time differs from panel to panel. Forced turnover is not a form of nonresponse. Rather, it is one of the criteria that determines who remains eligible for continued panel membership.

In contrast, so-called “normal” or unforced turnover is a form of nonresponse, in that people who are panel members and who have not reached the end of their eligibility leave the panel even though panel management might ideally like them to remain active members. The panel company may drop a member because he or she is not participating in enough surveys, may be continually providing data of questionable quality or engaging in some other forms of response that the panel’s users find objectionable. Panel members also may choose to opt out on their own for any number of reasons.

Many approaches could be used by those who strive to maintain panel membership among those members eligible to continue such membership. But there appears to be nothing reported in the research literature reporting the effectiveness of such possible approaches to reduce panel attrition or whether reducing attrition is desirable (obviously low attrition reduces new panel member acquisition costs which can make panel costs lower, but attrition that is too low may create its own difficulties, e.g., panel conditioning).

Those managing nonprobability panels can learn a great deal about attrition-related nonresponse by using the extensive data about their members gathered at the recruitment and profiling stages along with a host of other information that can be gleaned from past surveys the member may have completed. Analyses comparing those sampled members who do not drop out during their eligibility with those who do drop out due to reasons of normal turnover might lead to a clearer understanding of the reasons for the high rates of attrition. We are aware of no published accounts to verify that this is being done or what might be being learned.

RESPONSE METRICS

Callegaro and DiSogra (2008) point out that there currently are no widely-accepted metrics that can be used to accurately quantify or otherwise characterize the nonresponse that occurs at the recruitment stage for nonprobability online panels. This is because the base (denominator) against which the number of people who joined the panel (numerator) can be compared is often unknown. Furthermore, recruitment for many nonprobability online panels is a constant, ongoing endeavor. Thus, the concept of a recruitment response rate has a “moving target” aspect to it that precludes any standardization of its calculation. Although only sparsely documented in the literature, we believe that nonresponse at this stage is very high. At the profiling stage, a “profile rate” can be calculated which is similar to AAPOR’s Response Rate 6 (AAPOR RR6). The numerator is comprised of the number of those who completed profile questionnaires and possibly those who partially completed profile questionnaires. The denominator is comprised of the number of all people who initially enrolled in the panel, regardless of whether they started or completed the profiling questionnaire. The profiling rate is also an ever-changing number, since persons often are constantly coming into the panel over time. Thus, one could envision a profile rate being computed for set periods of time; e.g., a seven-day period, for many consecutive weeks. These rates could then be plotted to inform the panel managers how the profile rate is changing over time.

Thinking about the study-specific stage (where panel members are selected for participation in a specific study), Callegaro and DiSogra (2008) recommend several different rates that can be calculated:

- *Absorption rate* (Lozar, Manfreda, and Vehovar; 2002), which is the rate at which emails invitations sent by the panel managers to the panel membership actually reach the panel members. This is a function of the number of network-error undeliverable emails that are returned and the number of bounce-back undeliverable emails;
- *Completion rate*, which essentially is the AAPOR RR6 formula mentioned above, but limited to those panel members who are sampled for the specific survey;
- *Break-off rate*, which is the portion of specific survey questionnaires that were begun but never completed during the field period;

- *Screening completion rate* (Ezzati-Rice, Frankel, Hoaglin, Loft, Coronado, et al., 2000), which is the proportion of those panel members invited to participate in a specific survey, but who are deemed eligible or ineligible for the full questionnaire through a series of study-specific screening questions that they answer at the beginning of the questionnaire;
- *Eligibility rate*, which is the number of sampled panel members for a specific study that completed the screening and were found qualified compared to that same number plus the number who completed the screening and were found to be ineligible.

Finally, in terms of metrics that address panel maintenance, Callegaro and DiSogra (2008) suggest that the computation of the *attrition rate* be defined as “the percentage of [panel] members who drop out of the panel in a defined time period” (also see Clinton, 2001; and Sayles and Arens, 2007).

Of note, Bethlehem and Stoop (2007) point out that use of the term “response rate” in the context of a nonprobability panel survey can be misleading. Response rates can be boosted by pre-selecting the most cooperative panel members. This can invalidate comparisons with response rates from samples used with a different survey design. ISO 26362 (2009) recommends the use of the term *participation rate* rather than response rate because of the historical association of response rate with probability samples. The participation rate is defined as “the number of respondents who have provided a usable response divided by the total number of initial personal invitations requesting participation.”

COVERAGE ERRORS VERSUS NONRESPONSE BIAS

Errors of nonobservation are generally classified into either coverage or unit nonresponse errors (distinguished from item nonresponse errors). The section above on coverage errors notes that for most online panels there is no knowable target population or sampling frame that is well-defined. Without a sampling frame there can be no assessment of whether the frame covers the target population well—that is, whether those eligible for the sampling step are in some sense a microcosm of the full target population. Given the absence of a sampling frame in online panels, the conceptual difference between coverage error and nonresponse error also gets blurred, making it extremely difficult to ascribe errors in estimates to either source.

As discussed above, there are several mechanisms that can lead to unit nonresponse in online panel surveys. In this section we comment on the implications of nonresponse for survey quality, specifically the accuracy of survey estimates. Nonresponse may or may not result in biased estimates depending on (1) how likely a typical sample member is to participate (i.e., the average response likelihood, also known as response propensity) and (2) the relationship between the survey measure and response behavior (Bethlehem, 2002; Lessler and Kalsbeek, 1992).

The first factor, average response propensity, is a continuing and increasingly serious problem for all survey modes and samples, and online panels appear to be no exception. As occurs even with RDD samples, response rates in online panel surveys can vary dramatically, depending on the nature of the sample, the topic, the incentives, and other factors. Although response rate alone may be a poor indicator of error (Curtin, Presser, and Singer, 2005; Groves, 2006; Keeter, Miller, Kohut, Groves, and Presser, 2000; Merkle and Edelman, 2002), low response rates typically signal an increased concern for the possibility of nonresponse bias.

Research in the U.S. demonstrated only a weak association between response rate and nonresponse bias (Groves, 2006; Keeter, Kennedy, Dimock, Best, and Craighill, 2006). Additional research in the Netherlands has replicated these findings in online panels. One Dutch online panel comparison study (Vonk, van Ossenbruggen, and Willems, 2006) featured a common survey fielded independently by 19 different panels in the Netherlands. The study-specific response rates ranged from 18 percent to 77 percent, with an overall completion rate of 50 percent. The investigators found no meaningful differences between point estimates from the surveys with low response rates and those with high response rates. Similar results are reported by Yeager, Krosnick, Chang, Javitz, Levindusky, Simpser, and Wang (2009), although there was considerable variation between panels in terms of which estimates were closer to the “true” value. In other words, some panels were not reliably more accurate than others.

The second factor that is a concern with any survey, no matter what the mode or sample source, is the relationship between the survey measure (responses on the questionnaire, e.g., attitude ratings) and response behavior. This relationship has two critical properties from the researcher’s perspective. First, it is specific to each survey measure. That is, the likelihood of participating may be strongly associated with one factor (e.g., level of interest in the survey topic) and only weakly related or unrelated with a different factor (e.g., product satisfaction, purchase likelihood, presidential approval). The other important property is that it is generally very difficult, if not impossible, to measure the extent of this relationship since it requires knowledge of each sample member’s value on the survey measure, but typically these values would be known only for those who responded to the specific survey. Because unit nonresponse may be very high with a nonprobability panel, the relationship between response behavior and survey measures may be incalculable. However, since a good deal is known about panel members there is at least the opportunity to characterize the differences between responders and nonresponders to a given survey. As far as we can tell, this type of analysis is seldom done.

One potential reason for nonresponse bias is differential interest in the topic influencing participation decisions in online surveys. Early on, some believed that online panel surveys would be immune from this effect because the decision to participate in panels is general and response rates on individual surveys were high (Bethlehem and Stoop, 2007). However, as response rates declined, concern increased that advertising the survey topic in invitations may induce bias (and ethical guidelines concerning respondent treatment indicate that generally respondents should be informed about the topic of the survey, the length of the survey, and the incentives for the survey).

Unexpectedly, a recent experiment on this topic suggests that topic interest may have little if any effect on participation decisions. Tourangeau and his colleagues (2009) conducted a repeated measures experiment on topic-induced nonresponse bias using a pooled sample from two different online panels. They found that membership in multiple panels and a high individual participation rate were strong predictors of response in a follow-up survey but interest in the topic was not a significant predictor. Additional empirically rigorous studies of this sort are needed to confirm this null effect pertaining to topic interest, but the only available evidence suggests that general online survey taking behavior may be more influential in participation decisions than are attitudes about the survey topic.

NONRESPONSE BIAS AT THE RECRUITMENT STAGE. Given that we currently have little empirical evidence about nonresponders at the panel recruitment stage, the only chance to estimate potential effects from non-cooperation on bias at this stage is from what can be surmised about how those who joined the panel are likely to differ along key variables of interest compared to those who did not choose to join. Whether selective sampling and/or weighting can “correct” for such bias is addressed in Section 6.

DIFFERENCES BETWEEN ONLINE PANEL RESPONDERS AND NONRESPONDERS IN SPECIFIC STUDIES.

In contrast to the lack of information we have about nonresponders at the panel recruitment stage, those managing online panels often have large amounts of data they can use to study possible nonresponse bias that may exist on key variables in specific surveys of their members. We remind the reader of the framework used in this report: The sampling frame for a specific study is the database of online panel members, not a theoretical list of all persons who use the Internet. This is important for understanding our definition of nonresponse error and how it differs from coverage error. *Coverage error* refers to error resulting from the failure of the sampling frame (the panel) to adequately include all members of the target population. *Nonresponse error* arises because not all persons sampled for a study actually complete the questionnaire.

The low response rates sometimes observed in specific studies using nonprobability panels may signal a general risk of nonresponse bias but, ultimately, this error operates at the level of the individual survey question. We believe it is the researcher's responsibility to assess the threat to key survey measures from nonresponse using the extensive data they have available about nonresponders, knowledge of the survey subject matter and what has been documented about nonresponse bias in the literature.